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Cite as: Phys. Plasmas **31**, 063304 (2024); doi: 10.1063/5.0203446
Submitted: 13 February 2024 · Accepted: 19 May 2024 ·
Published Online: 7 June 2024

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Note: This paper is part of the Special Collection: Papers from the 65th Annual Meeting of the APS Division of Plasma Physics.
Note: Paper UI1 2, Bull. Am. Phys. Soc. **68** (2023).
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ABSTRACT

It was recently shown that the use of Coulomb logarithms appropriate for bremsstrahlung radiation (rather than transport processes) along with corrections for the Langdon effect and ion screening reproduced measurements of collisional absorption in well-characterized underdense plasmas [D. Turnbull *et al.*, Phys. Rev. Lett. **130**, 145103 (2023)]. However, it was recognized at the time that the use of the standard absorption-reduction factor from Langdon's seminal paper was inconsistent with the use of Coulomb logarithms that are thermally averaged over a Maxwellian electron-velocity distribution function. A more accurate approach would be to average over the expected super-Gaussian distribution function while accounting for the Gaunt factor's velocity dependence, which somewhat mitigates the Langdon effect; however, at that time, this theory matched the data less well. This conflict is now eliminated with the additional insight that the ionization state of our mid-Z ion species (when present) was lower than had been assumed, as evidenced by the Thomson-scattering data and time-dependent Cretin simulations. We are now able to show that an improved treatment of the Langdon effect provides the best match to data. Otherwise, the prior conclusions remain unchanged. We also show an example of the substantial expected impact to the absorption rate in calculations of indirect-drive hohlraums.

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I. INTRODUCTION

Inverse bremsstrahlung (i.e., collisional) absorption (IBA) is the most important process coupling lasers to plasmas at intensities below the relativistic regime. This makes it the primary mechanism by which lasers drive and/or preheat targets in the direct-drive,[1] indirect-drive,[2] and magnetized-liner[3] approaches to inertial confinement fusion. Despite its paramount importance and the persistent lack of consensus regarding how to compute IBA, there have been very few attempts to make focused absorption measurements to constrain modeling.

One recent such attempt reached the following high-level conclusions.[4] First, Coulomb logarithms (which are related by a scalar multiplier to "Gaunt factors" in the bremsstrahlung literature) derived for bremsstrahlung radiation (rather than transport processes) should be used, which differ most notably in their respective maximum impact parameters. Specifically, it was shown that the thermally averaged Coulomb logarithms from Sommerfeld and Maue[5] (as fit by Pradler and Semmelrock[6]) or Oster[7] get much closer to data than, for example, Lee and More[8] or Dimonte and Daligault.[9] Second, the Langdon effect must be taken into account, and the previous work did so using the standard absorption-reduction factor from Langdon's seminal paper.[10] Finally, a screening correction factor adapted from Rozsnyai[11] was added because the models still overestimated absorption by about ≈ 10% unless it was included.

However, it was inconsistent to use traditional Coulomb logarithms—which are typically thermally averaged over a Maxwellian electron distribution function (EDF)—but then tack on Langdon's absorption-reduction factor $f_{\mathrm{L}}^{\mathrm{orig.}}$, which implicitly assumes that EDFs are actually super-Gaussian. Matte provided a simple formula to estimate the super-Gaussian exponent characterizing the bulk portion of the EDF, and recent work has consistently validated that formula both experimentally[12,13] and computationally.[14] The factor $f_{\mathrm{L}}^{\mathrm{orig.}}$ is simply the ratio of the super-Gaussian and Maxwellian number densities at

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zero velocity (for equal temperature) based on the assumption that absorption is mostly determined by low-velocity electrons, but it therefore neglects the Gaunt factor velocity dependence. Directly integrating the Gaunt factor over the expected super-Gaussian EDF should ostensibly be more accurate.[15] However, absorption was overestimated by ≈ 6 to 10% on average when using this approach, whereas the models were within 2% on average using the conventional $f_L^{orig.}$; therefore, it was not used.

Here, we present an important refinement to our understanding of the plasma conditions in those experiments, namely, the ionization state of the mid-Z ion species (when present) was lower than had been assumed on the basis of steady-state FLYCHK[16] calculations. This correction is supported by the Thomson-scattering data as well as time-dependent Cretin[17] simulations. With this new information, we are now able to show that the improved treatment of the Langdon effect[15] results in the best match to data. Otherwise, the previous conclusions remain intact, including the need for the screening correction term. To support the revised conclusions, we herein present the data in substantially more detail than was shown previously. We also comment on recent work using molecular dynamics (MD) simulations that reached the opposite conclusion regarding the appropriate maximum impact parameter within the Coulomb logarithm and use those simulations to bolster the claim that screening does in fact start to become important at densities above 1% of the critical density. Finally, we discuss the likely implications for indirect-drive inertial confinement fusion by showing the expected change to the absorption rate within the hohlraum interior.

II. EXPERIMENTAL METHODS

A. Setup

The experiments were performed at the University of Rochester Laboratory for Laser Energetics' OMEGA-60 laser facility using the laser-plasma interaction platform.[4,18] A supersonic gas jet fitted with a Mach-3, 1-mm-diam. nozzle released a narrow column of gas at target chamber center (TCC). The gas was ionized and preheated quasi-isotropically by 13 3ω beams at time $t = 0$, each with an energy ranging from 32 to 190 J on average (depending on the shot) in a 500-ps-square pulse shape. A 527-nm probe beam with energy ranging from 1 to 4 J in a 100-ps picket pulse was then injected from the P9 port into the preformed plasma at either $t = 300$ or $t = 600$ ps (during or after the heater beams, respectively).

The previous publication included 22 data shots.[4] As we shall see, the refined analysis presented here relies on the ability to extract information out of subtle features in the ion-acoustic wave spectrum. Only the highest quality data were amenable to such analysis, so, here, we downselect to just 15 shots in the configurations previously referred to as Nos. 2 and 3. On all these shots, the heaters were conditioned with "SG5" phase plates (714-μm-FWHM), whereas the probe was conditioned with a "CircSG200" phase plate (165-μm-FWHM) to sample uniform conditions through the center of the plasma. All beams were pointed to TCC and the nozzle exit was positioned 1.5 mm from TCC. Eight shots used the reflective Thomson-scattering collection optics with a field of view of 1.5 mm and scattering angle of 59.9°; the other seven used the refractive collection optics with a > 4 mm field of view and scattering angle of 63.44°. The gas was either pure hydrogen (five shots) or a mixture of 43% nitrogen and 57% hydrogen (ten shots). The input parameters for each shot will be detailed in Table I.

TABLE I. Summary of key shot parameters. The two gases are grouped together, but the shots are ordered chronologically otherwise. The ionization state for nitrogen listed here is determined or at least constrained by the Thomson-scattering data.

Shot 10-	Gas	P_{gas} (psi)	$E_{htr.}$ (J)	t_{probe} (ns)	Z_N	$A_{meas.}$ (%)
-2729	N_2/H_2	136	189	0.6	6.5	1.38
-2737	N_2/H_2	137	35	0.3	≤ 6(5)	3.5
-2738	N_2/H_2	135	189	0.3	≤ 6.5(6)	1.34
-4035	N_2/H_2	137	100	0.3	≤ 6(5)	1.88
-4037	N_2/H_2	135	100	0.6	6	2.37
-4041	N_2/H_2	136	32	0.6	6	3.91
-6315	N_2/H_2	146	101	0.6	6	2.58
-6316	N_2/H_2	72	101	0.6	5.5	0.97
-6317	N_2/H_2	38	95	0.6	5	0.47
-6319	N_2/H_2	268	97	0.6	≥ 6(7)	5.05
-2727	H_2	499	189	0.6		0.73
-4031	H_2	498	89	0.3		1.48
-6309	H_2	501	94	0.3		1.43
-6310	H_2	252	99	0.3		0.39
-6311	H_2	126	99	0.3		0.15

B. Absorption measurements

The transmission T of the probe beam through the plasma was determined by comparing an upstream pickoff of the beam just prior to the target chamber to a downstream measurement after TCC using the P9 transmitted beam diagnostic (TBD).[19] The ratio of these detectors was calibrated daily for each shot using shots with no gas injected at TCC. Data shots then differed only by the presence of the plasma. By maintaining backscatter losses below the noise level of ≈ 0.1% and remaining below threshold for any beam spray[20] that could result in unmeasured transmission off the edge of the $f/4.2$ TBD scatter plate, the path-integrated absorption was simply $A = 1 - T$. Figure 1 shows the setup and examples of both the calibration and shot data. Note that the near field of the beam transmitted through plasma is highly modulated compared to the beam propagating through vacuum, with a larger peak-to-valley amplitude, but saturation was avoided, and we verified that the camera's response was highly linear over its entire range. The absolute measurement uncertainty was estimated to be $(-0.12\%, +0.07\%)$.

C. Thomson scattering

The imaging Thomson-scattering system[21] was the other key diagnostic from which spatially resolved plasma conditions were determined to use as inputs to the absorption calculations. A first example is shown in Fig. 2. This shot featured the lowest density used for the nitrogen–hydrogen mixture. While this resulted in a very low measured absorption of just 0.47% (where unfortunately the experimental uncertainty is relatively large), the lower density is preferable for Thomson scattering because electron-plasma wave (EPW) features are less collective, i.e., more sensitive to the EDF.

The properties of ion-acoustic waves (IAWs) in multi-species plasmas were described by Williams for the case of a C–H mixture,[22] which also applies to our N–H case. In short, IAW peaks are asymmetric because they consist of both a slow acoustic mode (with frequency

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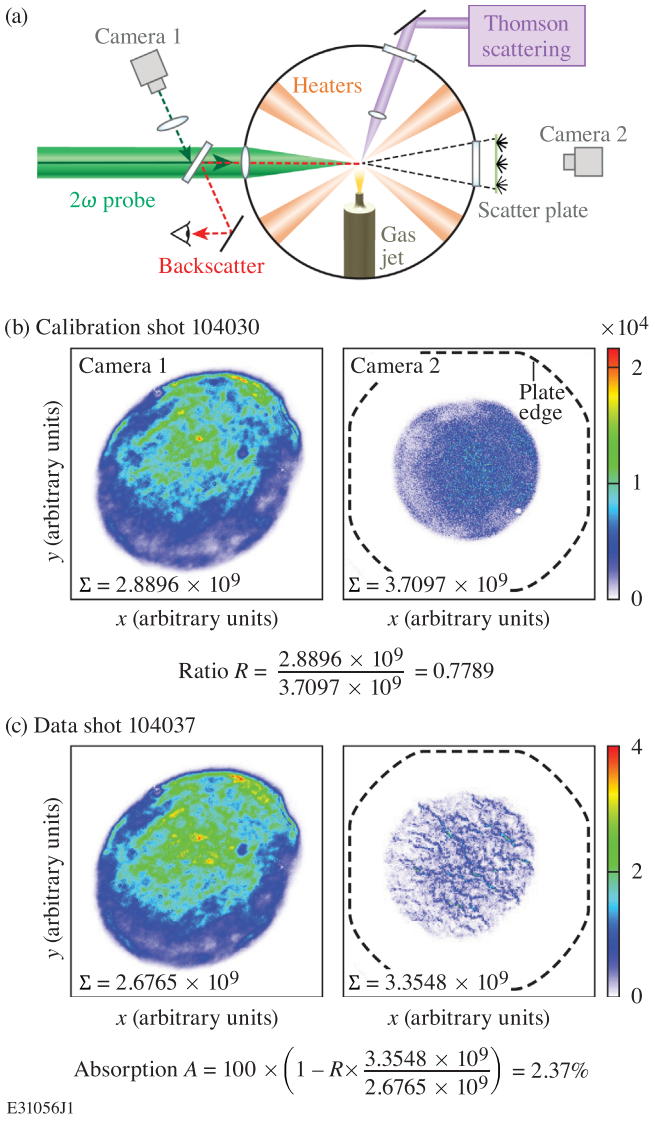


FIG. 1. (a) Setup for the absorption platform. (b) Calibration shot example through vacuum. (c) Data shot through plasma with inferred absorption.

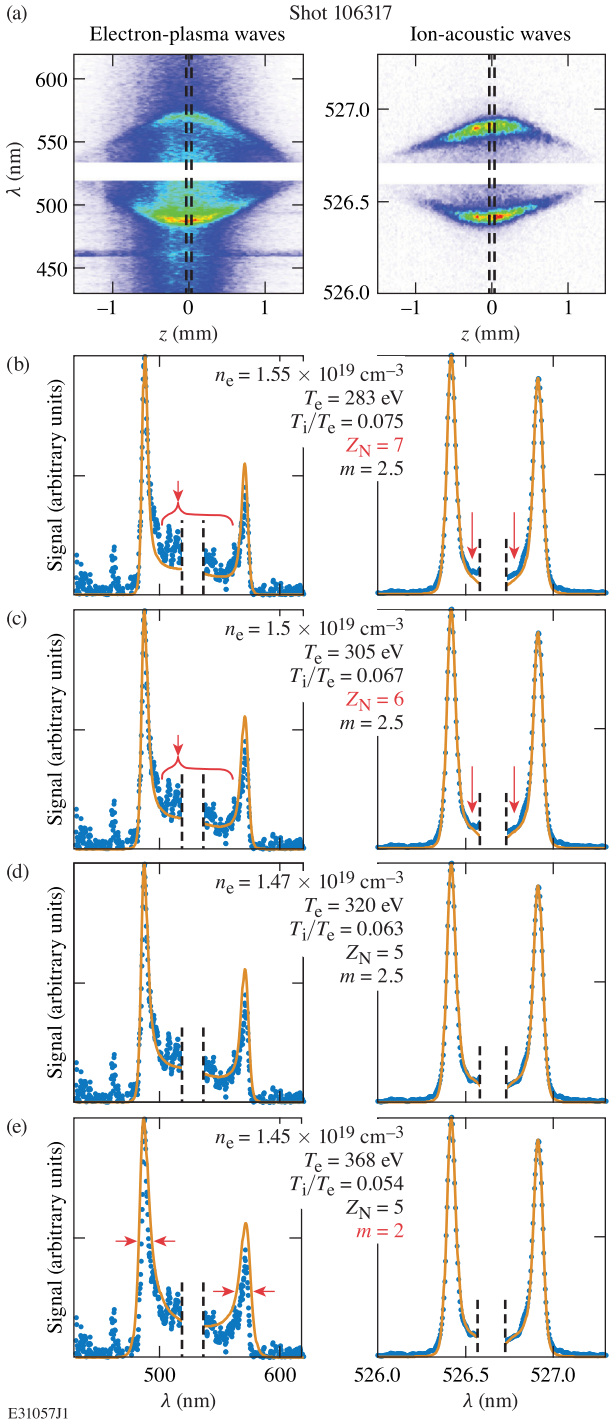


FIG. 2. (a) Raw data for both the electron-plasma wave (EPW) features and ion-acoustic wave (IAW) features. (b) Attempting to fit with $Z = 7$ for the nitrogen ions, the shoulder on the IAW peak from the slow mode is poorly reproduced, as is the plateau region in the EPW feature. (c) With $Z = 6$, there is improvement, but residual discrepancy on both the aforementioned features. (d) $Z = 5$ results in the best fit overall. (e) With a Maxwellian rather than super-Gaussian assumption for the electron distribution function, the EPW peak widths and shape are highly inconsistent.

less than the average-ion sound speed) and a fast acoustic mode (with frequency greater than the average-ion sound speed). Differences in the damping rates of these modes affect their relative prominence. For our conditions, the fast mode is more weakly damped and produces the strong peak, with position affected by the ratio of hydrogen's ionization state to its mass (i.e., 1). The slow mode therefore shows up as a shoulder on the interior side of each IAW peak that is sensitive to the ratio of nitrogen's ionization state to its mass (which is variable).

Figure 2(b) shows that this shoulder feature in the IAW spectrum does not fit well for shot 106317 when assuming $Z = 7$ for the nitrogen ions. With $Z = 6$ [Fig. 2(c)], there is some improvement, but the best fit is found assuming $Z = 5$ as shown in Fig. 2(d). This represents a substantial revision to our understanding of the plasma conditions,

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because we had previously assumed that nitrogen was fully ionized at the density peak in all relevant cases on the basis of steady-state FLYCHK[16] calculations. Note that with a lower ionization state for nitrogen, the inferred temperature is higher to retain agreement with the IAW peak locations, and, in turn, this requires a lower electron density to preserve a good match to the EPW peak locations. All of these changes will result in lower predicted absorption.

As in the previous summary of this work, the bulk portion of the laser-heated EDF that impacts absorption is assumed to conform to a super-Gaussian distribution $f_e(v)$ in the form

$$f_e(v) = c_m \exp\left(-\left(\frac{v}{v_m}\right)^m\right), \quad (1)$$

where $c_m = \frac{n_e}{4\pi}\frac{m}{\Gamma(3/m)v_m^3}$, $v_m = \sqrt{\frac{3T_e}{m_e}\frac{\Gamma(3/m)}{\Gamma(5/m)}}$, Γ is the gamma function, and the exponent m was given by Matte *et al.*,[23]

$$m(\alpha_L) = 2 + 3/(1 + 1.66/\alpha_L^{0.724}), \quad (2)$$

where α_L is the Langdon parameter,[10]

$$\alpha_L = Z_{eff} v_o^2/v_t^2, \quad (3)$$

in which $Z_{eff} = \langle Z^2\rangle/\langle Z\rangle$, $v_o = eE/m_e\omega$ is the oscillatory velocity of electrons in the laser field, and $v_t = \sqrt{T_e e/m_e}$ is the electron thermal velocity (SI units used throughout except for electron temperatures in eV).

Recall that the EPWs are sensitive to the EDF at this low density. The increase in the inferred electron temperature that accompanies the reduction in the assumed ionization state raises the plateau region inside of the EPW peaks, resulting in a better overall match to the EPW features as well. The super-Gaussian exponent of $m = 2.5$ used in Figs. 2(b)–2(d) is consistent with Eq. (2) using the plasma conditions from Fig. 2(d) and the laser intensity on that shot [see Fig. 11(c) in Appendix A]. Figure 2(e) validates the super-Gaussian assumption by showing that the EPW peak widths and shapes are highly inconsistent with a Maxwellian assumption (i.e., $m = 2$). We should keep in mind, however, that Eq. (2) was not derived from first principles; rather, it was determined from fits to the bulk electron distribution functions observed in Fokker–Planck simulations over a range of intensities.[23] However, the fact that Eq. (2) seems to work well is consistent with prior observations.[12,13]

Figure 3 shows another example—shot 106315—in which the Thomson-scattering IAW features constrained the ionization state of nitrogen. On this shot, the requested gas jet backpressure was much higher, yielding a peak electron density of $n_e = 7.35 \times 10^{19}$ cm^{-3} (almost 5× larger than the previous example), which makes it more representative of the other shots in this database. Density is the strongest lever with which to control the electron temperature in these gas-jet plasmas, with higher density resulting in more absorption and therefore higher temperatures. The inferred electron temperature for shot 106315 is about 2× larger than for the previous shot 106317. Unsurprisingly, the data suggest that a higher ionization state of $Z = 6$ is reached.

A summary of the inferred ionization states for each shot that included nitrogen ions is presented in Table I. The shots in which the data seemed to clearly discriminate between ionization levels were typically those on which the probe was timed later. The shots with earlier timing were often more ambiguous, which may be due to the fact that

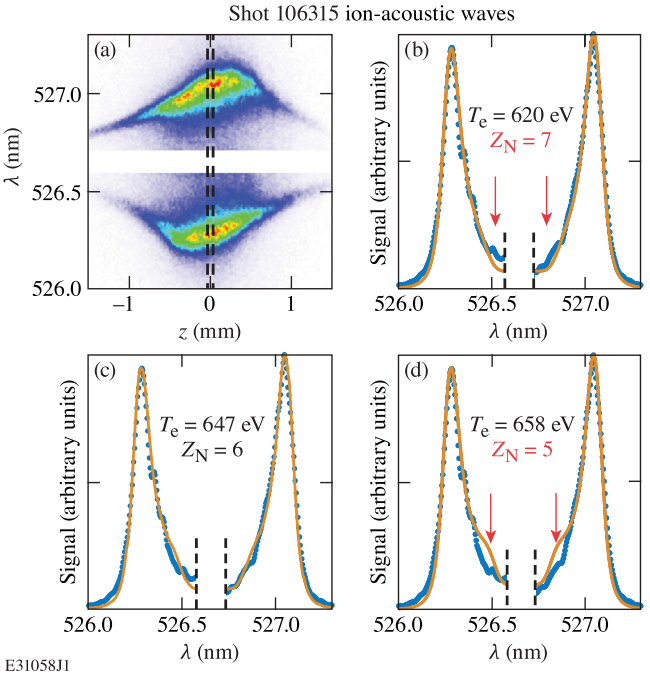


FIG. 3. (a) The ion-acoustic wave (IAW) data. (b) With $Z = 7$, the fit is below the slow-mode shoulder in the data. (c) $Z = 6$ produces the best overall fit. (d) With $Z = 5$, the fit is above the slow-mode shoulder.

the plasma conditions were changing more rapidly. In those cases, we use their late-time companion shots (with similar parameters aside from probe timing) to at least bound the ionization state (assuming it was increasing monotonically with time), which is listed in the table along with the parenthetical value used in the absorption calculations. Since it is relatively cumbersome to determine the ionization state with high confidence at every position along the probe's propagation path, the steady-state ionization profiles used previously were scaled using the new values at the peak density location. This may be somewhat inaccurate in the low-density wings, but these regions do not contribute much to the overall absorption (see the transmission curves in Figs. 10–12). The spatially resolved plasma conditions for all of the shots are detailed in Appendix A.

The key point to note in Table I is that the ionization trends appear to be very sensible. From shot 106317 to shot 106316 and then shot 104037 with its nominal repeat shot 106315, the ionization level increased with density (at a fixed average heater-beam energy). We assume that trend continued to shot 106319 and use the lower-pressure shots to bound its ionization state. From shots 104037 and 106315 to shot 102727, we see a weaker trend with heater-beam energy, but again consistent with what we would expect—more heating results in higher ionization states. The uncertainty in Z is taken to be ±0.25 for the shots with constraining, high-quality Thomson data, and ±0.5 for the shots that are merely bounded by their companions.

D. Cretin simulations

As a further sanity check to verify that the ionization states inferred from Thomson scattering are not entirely unexpected,

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time-dependent simulations were performed using the collisional-radiative code, Cretin.[17] Only the peak density was modeled using 0D simulations; Cretin can also be run inline as part of a radiation-hydrodynamics code that would better account for plasma expansion and heat transport, but this is left for future work. Shot 106317 (featured in Fig. 2) was simulated using the laser intensity profile shown in Fig. 4(a) along with an appropriate initial density of the nitrogen-hydrogen neutral gas mixture to match the electron density at the time of the measurement, as shown in Fig. 4(b). The simulation slightly underestimates the observed electron temperature [Fig. 4(c)] and overestimates the ionization state determined from the Thomson-scattering data [Fig. 4(d)]. However, the key takeaway is that while the first five electrons are stripped quite quickly, the timescales for reaching N^{6+} and N^{7+} can be slow compared to the experiment. This bolsters the case for using the lower ionization states listed in Table I in the absorption calculations instead of the steady-state values assumed previously.

III. ABSORPTION CALCULATIONS

We can then use the spatially resolved plasma conditions to estimate how much absorption should have occurred as the probe beam propagated through the plasma. The standard absorption rate (with units m^{-1}), corrected by f_L to account for the Langdon effect (as is somewhat common), as well as by (f_{sc}) to account for ion screening (as is less common) is

$$\kappa = \nu_{ei}\frac{n_e/n_c}{c\sqrt{1-n_e/n_c}} f_L f_{sc}, \tag{4}$$

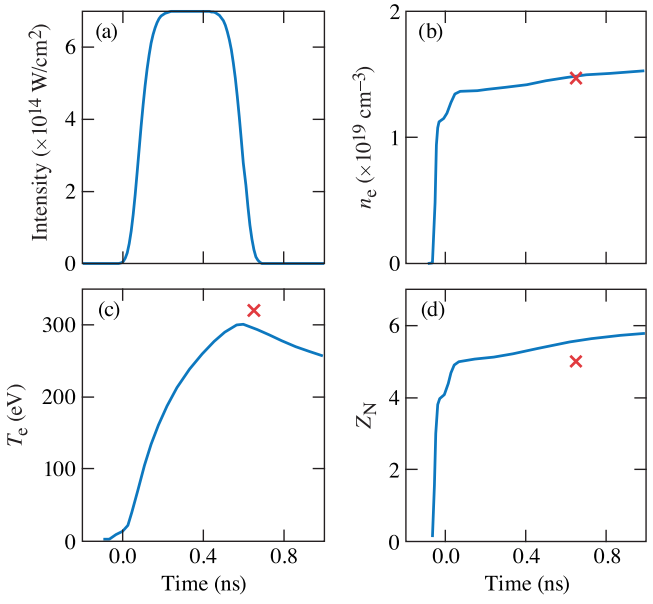


FIG. 4. The simulations were initialized using the laser intensity (a) as well as an initial neutral–gas mixture density chosen to match the observed electron density at the time of the measurement (denoted by the x), as shown in (b). Similarly, Cretin's predicted temperature and ionization-state time histories are compared to the respective measurements in (c) and (d).

where $\nu_{ei} = 2.91 \times 10^{-12} T_e^{-3/2} \sum_i (Z_i^2 n_i \ln \Lambda_{IB,i})$ is the electron–ion collision rate including the summation over all ion species.

A. Coulomb logarithms

The Coulomb logarithm $\ln \Lambda_{IB,i}$ for each ion species generally arises from the need to bound an integral in the classical derivation for the electron–ion collision rate using maximum and minimum impact parameters b_{max} and b_{min}, but one or both are ostensibly constrained depending on the derivation. The previous summary[4] of these results demonstrated that Coulomb logarithms derived for transport processes[8,9]—which typically feature a Debye length as the maximum impact parameter—should not be used for IBA. We will reiterate that conclusion later but here downselect to only the more accurate bremsstrahlung theories derived from binary collision models,

$$\text{Sommerfeld and Maue}:^{5,6}\ \frac{\pi}{\sqrt{3}}\langle g_{ff}\rangle[f_e(v),\omega,Z], \tag{5}$$

$$\text{and Oster}:^{7}\ \ln\left(\left(\frac{2}{e^{\gamma}}\right)^{5/2}\frac{v_t/\omega}{\max\left(\frac{Ze}{4\pi\varepsilon_0 T_e},\frac{\hbar}{1.68\sqrt{m_e T_e e}}\right)}\right). \tag{6}$$

Sommerfeld's seminal result—a function of the EDF $f_e(v)$, laser frequency ω, and Z—is an exact quantum-mechanical solution for the bremsstrahlung emitted in a single binary electron–ion collision, but it contains complicated hypergeometric functions that can be computationally prohibitive to evaluate.[5] However, one can alternatively use tabulated results[24] or one of the many attempts at constructing a fit over some range of parameter space.[6,25–28] To estimate Sommerfeld in our analysis, we use Pradler's fitting formula,[6] which is summarized here to make this work more self-contained. Starting with the Sommerfeld parameter $\nu_i = \alpha Z c/v_i$ and the emitted energy fraction $x = 2\hbar\omega/(m_e v_i^2)$, where $\alpha \approx 1/137$ is the fine-structure constant (not to be confused with the Langdon parameter denoted by α_L that was introduced earlier), c is the speed of light, v_i is the initial nonrelativistic electron velocity, and $\hbar$ is Planck's constant, we have the velocity-dependent Gaunt factor

$$g_{ff}(\nu_i,x) \approx \frac{\sqrt{3}}{\pi}\left[\ln\left(1+\frac{1}{\nu_i e^{\gamma}}\frac{1+\sqrt{1-x}}{1-\sqrt{1-x}}\right) + \frac{\pi^2\nu_i^2}{\sinh^2(\pi\nu_i)}\ln(\nu_i e^{\gamma}) - \frac{10}{3}\nu_i^2 e^{-\frac{3}{2}\pi\nu_i} \times\left(\frac{5}{1+e^{10(x-6/7)}}-2\right)\right] + \frac{x\nu_i}{\pi^{-2/3}+x\nu_i}, \tag{7}$$

where $\gamma \approx 0.577$ is Euler's constant. Setting aside for the moment the fact that the EDF is actually super-Gaussian rather than Maxwellian, we follow the usual convention and integrate over a Maxwellian to get the thermally averaged Coulomb logarithm. Introducing the dimensionless energy $\xi = \hbar\omega/(T_e e)$, the most probable value of ν_i, $\nu_p = \alpha Z c\sqrt{m_e/(2T_e e)}$, and the dimensionless initial energy $u = m_e v_i^2/(2T_e e)$, we have the following:

$$\langle g_{ff}\rangle = \int_{\xi}^{\infty} g_{ff}\left(x=\frac{\xi}{u},\nu_i=\frac{\nu_p}{\sqrt{u}}\right)e^{-u}du. \tag{8}$$

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Oster [Eq. (6)] was also derived from a binary-collision approach, but additional assumptions (straightline approximation to the hyperbolic trajectory) facilitated simpler formulas.[7] Equation (6) combines their results from classical and quantum-mechanical approaches in a way that isolates the common maximum impact parameter $b_{max} = v_t/\omega$ from the minimum impact parameter that is either the distance of closest approach (classical) or something close to the thermal de Broglie wavelength (quantum-mechanical). In the quantum limit, the Coulomb logarithm simplifies considerably to $\ln(4T_e e/\hbar\omega e^{\gamma})$, which is found elsewhere.[29–31] Though not shown, the well-known formula for the high-frequency limit (when $\omega \gg \omega_p$, where ω_p is the plasma frequency) from Dawson and Oberman,[32] later revised by Johnston and Dawson[33] and adopted by the NRL plasma formulary,[34] has the same format as Oster but lacks the order-unity terms to reflect the indeterminacy of b_{min} in that derivation, although b_{max} was rigorously derived. They could instead have chosen minimum impact parameters in order to match the results of Oster. The justification offered in the bremsstrahlung literature for this common b_{max} is that the duration of the collision must be short compared to the period of the wave, otherwise the heating is rendered ineffective.

Figure 5 shows calculations comparing an explicit evaluation of Sommerfeld and Maue[5] to the fit from Pradler and Semmelrock[6] [Eqs. (5), (7), and (8)] or the alternative simpler formula from Oster[7] [Eq. (6)] for low- and mid-Z plasmas at a wavelength of 526.7 nm. Pradler's fit provides a good approximation to Sommerfeld over the entire range of temperatures for both cases. Oster, on the other hand, performs very well for hydrogen at temperatures above 100 eV and is reasonably accurate for the case of $Z = 10$ over the range of about $500 < T_e < 5000$ eV, but the error is unacceptably high outside of this range, and this may limit the utility of the simpler formulas. Note that Oster actually becomes negative at temperatures below 20 eV. This is nonsense, of course, but does not mean that Oster is invalid generally; rather, the conventional Coulomb logarithms[7,32–34] require that $b_{max} \gg b_{min}$, which is not satisfied for these cold temperatures. Some codes avoid negative heating rates by using a form like $0.5\ln(1+\Lambda^2)$ instead of $\ln(\Lambda)$, but this treatment would not significantly reduce the error between Sommerfeld and Oster at low temperatures. We will revisit this point later as it pertains to recent MD simulations.[35]

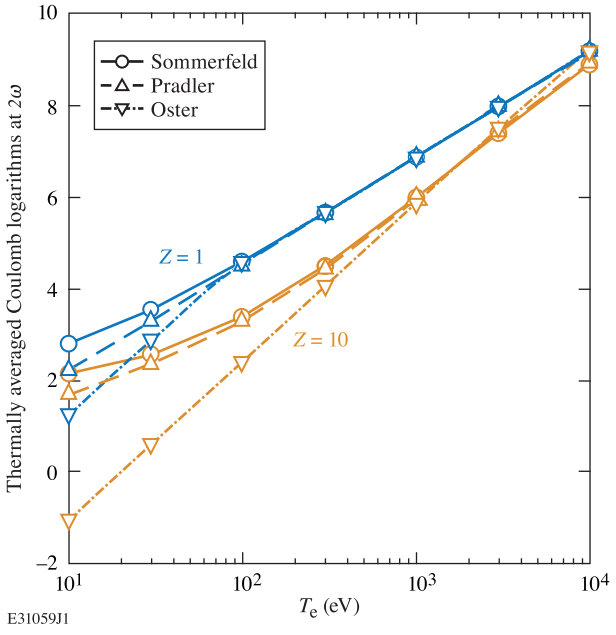


FIG. 5. Comparing the various Coulomb logarithm formulas for low- and mid-Z plasmas, a wavelength of 526.7 nm, and temperatures between 10 eV and 10 keV.

B. Langdon effect

Langdon showed[10] that IBA preferentially heats low-energy electrons, which distorts the EDF away from a Maxwellian and toward a higher-order super-Gaussian [in the form of Eq. (1)] up to a limit of $m = 5$. Light then couples less efficiently to the distorted EDF because of the smaller number of low-velocity electrons, which reduces absorption. The magnitude of this effect is governed by the ratio [Eq. (3)] of the electron–ion collision rate (governing the absorption) to the rate of electron–electron collisions that are acting to restore a Maxwellian EDF.

The original absorption-reduction factor in Langdon's seminal paper was

$$f_L^{orig.} = 1 - 0.553/\left[1 + (0.27/\alpha_L)^{0.75}\right]. \tag{9}$$

As mentioned earlier, however, this factor is simply the ratio of the expected super-Gaussian EDF [which was later found[23] to be predicted by Eq. (2)] at zero velocity to a Maxwellian EDF of the same temperature, $f_{SG}(v=0)/f_{m=2}(v=0)$, based on an assumption that absorption is mostly determined by the number of low-velocity electrons.

However, we now have a glaring inconsistency. The Coulomb logarithms were just thermally averaged over an EDF that was assumed to be Maxwellian, but by tacking on the Langdon factor, we are implicitly assuming that the EDF is actually super-Gaussian. Directly integrating the velocity-dependent Gaunt factor [e.g., Eq. (7)] over the expected super-Gaussian EDF is a more streamlined approach that should be more accurate.[15] Retaining the usual convention, however, that the Coulomb logarithms are derived for a Maxwellian EDF, and the Langdon factor is applied as a correction (or not, based on user discretion), we can define a new Langdon correction factor that effectively undoes the integration over a Maxwellian and replaces it with an integration over a super-Gaussian EDF with exponent $m(\alpha_L)$,

$$f_L^{new} = \frac{m}{2}\frac{v_{m=2}^2}{v_m^m}\frac{c_m}{c_{m=2}}\frac{\int_0^\infty g_{ff}(v)\exp(-(v/v_m)^m)v^{m-1}dv}{\int_0^\infty g_{ff}(v)\exp(-(v/v_{m=2})^2)vdv}, \tag{10}$$

where c_m and v_m were defined earlier after Eq. (1), and for the Gaunt factor, we again use Eq. (7). This improved treatment of the Langdon effect is covered more thoroughly by Sherlock *et al.* using a different approach that yields comparable results.[15] Note that this form of thermal averaging neglects stimulated emission, which is negligible for the conditions of the experiments.

Figure 6 shows a few examples of how Eq. (10) can differ significantly from Eq. (9) depending upon the specific plasma conditions. Figure 6(b) shows that the ratio of the two factors—which is effectively the fractional increase in the absorption rate when switching from the old theory to the new theory—can be as large as ≈ 30%. Note that Eq. (10) provides a unique absorption-reduction factor for each ion species, so they should be applied within the summation term in Eq. (4) rather than as one single factor that impacts each species equally.

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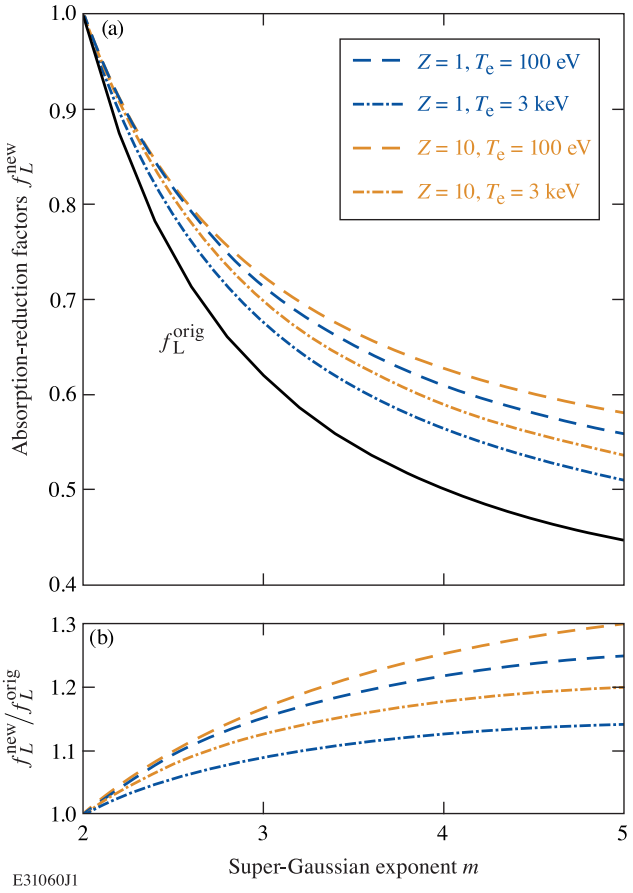


FIG. 6. (a) Comparing Langdon's original absorption-reduction factor to results from a proper integration over the super-Gaussian electron distribution function for various cases. (b) The ratio of the new factor to the old factor for each case.

C. Screening

Although the Coulomb logarithms introduced in Sec. III A were derived from a binary collision model (single electron interacting with a single ion) and thus neglected any medium effects, we can conclude from Dawson's work that medium effects are not important in the high-frequency limit.[32,36] Nonetheless, we shall see that both these experiments and MD simulations[35] suggest that ion screening becomes important for densities of interest above $n_e \gtrsim 0.01 n_c$. To account for this, we define the screening correction factor,

$$f_{sc} = \frac{\ln\left(\frac{2T_e e}{\hbar\omega}\frac{y}{\sqrt{1+y^2}}\right) - \frac{1}{2(1+y^2)}}{\ln\left(\frac{2T_e e}{\hbar\omega}\right)}, \tag{11}$$

where $y = l_{sc}/b_{max} = l_{sc}\omega/v_t$ is the ratio of the effective screening length to the maximum impact parameter. This term (which is also explained more thoroughly by Sherlock *et al.*[15]) is Rozsnyai's simplified Gaunt factor[11] normalized by the Gaunt factor derived in the Born approximation without screening, and is valid for $y \geq 0.005$ and $\hbar\omega \ll T_e e$, which applies to most laser-produced plasmas. Although the Debye length is the most common choice for the screening length in weakly coupled plasmas, we choose instead the mean inter-ionic distance, or Wigner–Seitz radius, $l_{sc} = a_i = (3/4\pi n_i)^{1/3}$, where n_i is the ion density, in order to match the data. Our qualitative explanation for the impact of screening in the case when $a_i < b_{max}$ is that electrons interact with more than one ion simultaneously, resulting in a cancelation of the electric fields and therefore less perturbation to the electron's trajectory in the laser field. [With this picture in mind, note that for our multi-species mixture of nitrogen and hydrogen in which the nitrogen ions dominated the absorption—$Z_N^2 n_N \ln\Lambda_{IB,N} \gg Z_H^2 n_H \ln\Lambda_{IB,H}$—$n_N$ was used in Eq. (11) because only another nitrogen ion with comparable charge state would effectively screen a nitrogen ion.] The choice of $f_{sc}(a_i)$ we offer as the best fit to the data should motivate further MD simulations and theoretical analysis for weakly coupled plasmas to more accurately determine the effect of close ion spacing.

IV. RESULTS

Here, we refine (but not replace) the conclusions that were published previously.[4] Figure 7 compares predicted to measured absorption for the 15 shots using Pradler's fit to Sommerfeld for the Coulomb logarithm but varying other model components. It is helpful to separate the pure-H_2 shots (plotted in blue) from the N_2/H_2 shots (plotted in orange), because the mixture shots were impacted to a greater degree by the Langdon effect, whereas the hydrogen shots were

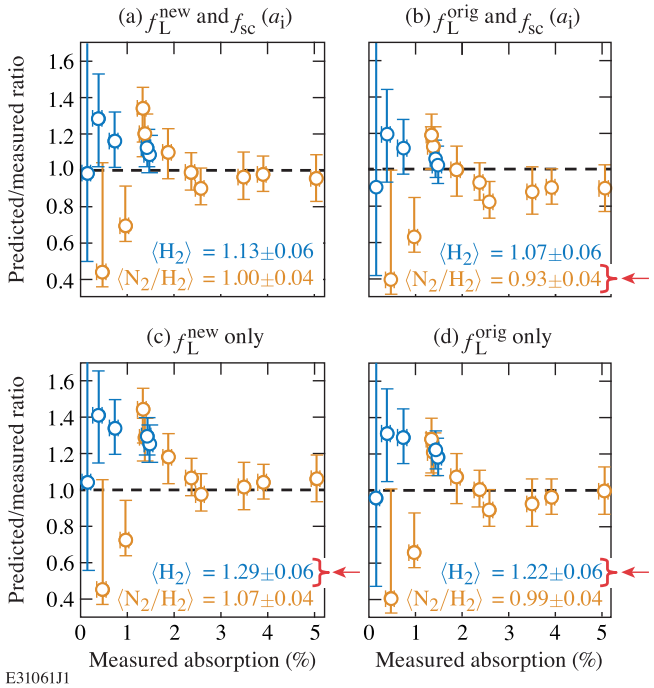


FIG. 7. Comparing predicted to measured absorption using the Sommerfeld Coulomb logarithm in each case and either (a) the new Langdon factor and screening, (b) the original Langdon factor and screening, (c) the new Langdon factor but no screening, or (d) the original Langdon factor with no screening. In each subfigure, the labels indicate the average and standard deviation of the predicted-to-measured ratio for each set of shots using that model.

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affected more strongly by ion screening (n_i being much higher to achieve comparable n_e for $Z = 1$).

Figure 7(a) uses both the improved treatment for the Langdon effect [Eq. (10)] and the screening correction [Eq. (11)] and results in the overall best match to data. It is particularly satisfying to see all of the mixture shots with measured absorption $> 1.5\%$ (which are therefore the least affected by the absolute measurement uncertainty) all predicted to better than 10% and well within the uncertainty. Several of these shots were outliers in the previous publication[4] prior to updating the ionization states in the calculations.

Figures 7(c) and 7(d) highlight the significant impact that the ion screening term has on the hydrogen shots in particular. Without screening, the predictions are higher by 16% on average, which significantly widens the gap between the predictions and the measurements. With ion screening [Fig. 7(a)], the overestimate is reduced to about the level of the uncertainty.

If we include screening but revert to using Langdon's original absorption-reduction factor [Eq. (9)], Fig. 7(b) shows that the predictions are lower by an additional 7% on average, resulting in systematic overestimates for the highest-quality N_2/H_2-mixture data.

In the earlier publication,[4] a key conclusion was that Coulomb logarithms derived for transport processes—which typically use a Debye length for the maximum impact parameter—are wholly inappropriate for IBA calculations. Dimonte's formula[9] was used as an example, which features the electron-only Debye length that is larger than v_t/ω by the ratio $\omega/\omega_p = \sqrt{n_c/n_e} \approx 7$ for most of these shots, thus representing a substantial change in the calculations. Since the use of the Debye length implies screening is already being taken into account, we use no additional screening correction, in which case the weighted average ratio of predicted to measured absorption for all of the shots is 1.48 ± 0.04—a significant disagreement that once again invalidates the use of such theories. It was shown that the widely used Lee and More formula[8] resulted in similar disagreement.[4] Therefore, we reiterate that Coulomb logarithms for IBA should depend on the laser frequency rather than the plasma frequency.

V. DISCUSSION

A. A comment on recent MD simulations

Given that conclusion, it is worth commenting on a recent publication that used MD simulations to justify reaching the opposite conclusion, namely, that plasma-frequency rather than laser-frequency dependence in b_{max} is appropriate. Their approach was to fit the observed heating rates from MD simulations of IBA using a conventional $\ln(b_{max}/b_{min})$ form [similar to Eq. (6)] while testing both options for the frequency dependence in b_{max}. Plasma-frequency dependence appeared to work best, but they matched simulations only with the help of a separate multiplier outside the Coulomb logarithm (unjustified by any theory) that had to be tuned between datasets.

This erroneous conclusion was simply an artifact of their choice of $T_e = 10$ eV in the simulations. As shown in Fig. 5, the conventional form of the Coulomb logarithm[7,32–34] breaks down when $b_{max} \approx b_{min}$, which occurs at very low temperatures. The exact result of Sommerfeld, however, can handle that situation. Restricting ourselves to a subset of their simulation results for which $v_o \ll v_t$ so we can ignore the Silin effect,[37,38] we plot in Fig. 8 the expected heating rate using an explicit evaluation of Sommerfeld for their laser and plasma conditions against the results of their simulations.

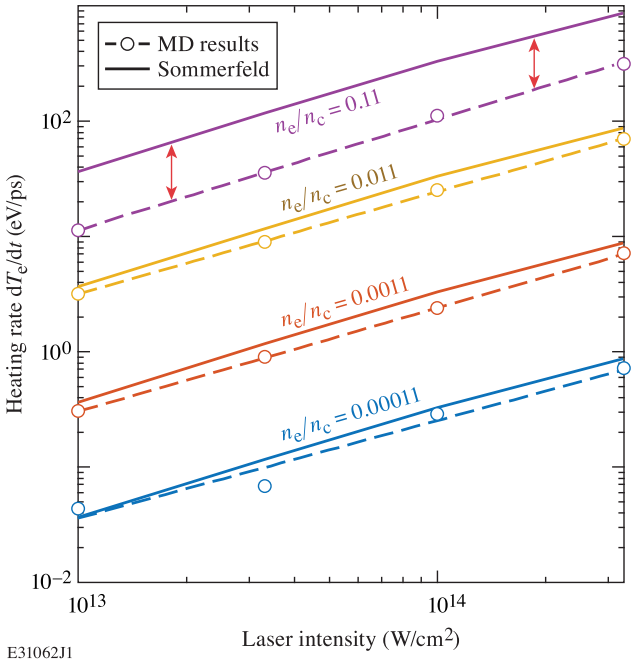


FIG. 8. Comparing predictions using an explicit evaluation of Sommerfeld for the Coulomb logarithm to the heating rates observed in recent MD simulations.[35]

For densities $n_e/n_c \leq 0.011$, the predictions are in quite good agreement with the MD simulations, overestimating the heating rate by just $18 \pm 14\%$ on average. Note that there is just about one decade in between the simulated heating rates each time the plasma density is increased by one order of magnitude, which is consistent with the linear density dependence in the heating rate *outside* the Coulomb logarithm, and therefore lack of any density dependence (i.e., plasma-frequency dependence) *inside* the logarithm, contrary to their conclusions.

However, this trend does break down for the case of $n_e/n_c = 0.11$, for which Sommerfeld overestimates the heating rate by $67 \pm 3\%$, which we take to be additional evidence that screening does indeed become important in between 1.1% and 11% of critical density. This is consistent with our conclusions from experiments, although the screening correction factor [Eq. (11)] does not capture the more severe reduction in heating rate observed in the simulations, even using the most restrictive Wigner–Seitz screening length. This may be due to the fact that the cold temperature once again violates an assumption, namely, that $\hbar\omega \ll T_e e$, which is required for the validity of Eq. (11).

B. Implications for inertial confinement fusion

The previous paper estimated the impact of the proposed model changes to the absorption rate in the coronal plasma of a directly driven inertial confinement fusion implosion. Since the refined theory presented here mostly concerns the treatment of the Langdon effect, which is a relatively modest correction factor for typical direct-drive conditions, the implications for that case are expected to be largely unchanged.

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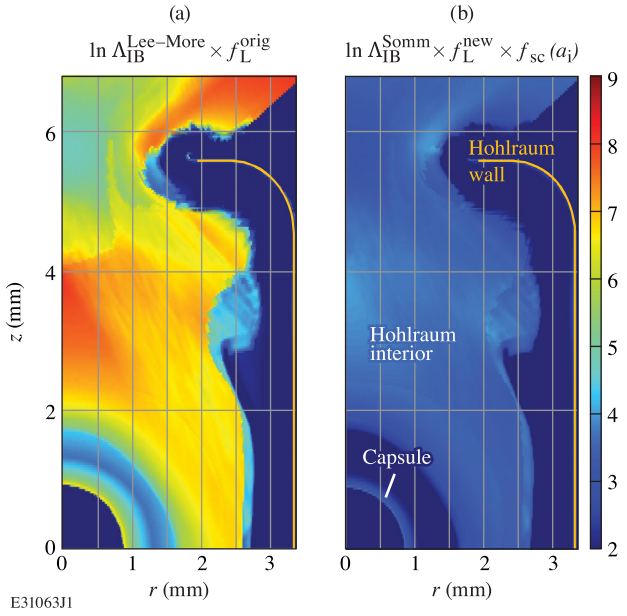


FIG. 9. (a) A map of the product of the Lee–More Coulomb logarithm and original Langdon correction factor (together comprising the old default model) in an indirect-drive hohlraum. (b) Keeping the plasma conditions fixed, the new model (Sommerfeld Coulomb logarithm along with the new Langdon factor and the ion-screening correction) would nearly halve the absorption rate.

Here, we estimate the expected impact for an indirectly driven implosion using a hohlraum platform. Plasma conditions were extracted from a radiation-hydrodynamic simulation performed using HYDRA.[39] Although the details are not believed to be important, and we expect similar differences in other implosion designs, the particular snapshot that we will use is at $t = 11$ ns during peak power for a high-foot implosion using a CH capsule in a 6.72-mm-diam. depleted-uranium hohlraum with an intermediate 0.6 mg/cc helium gas fill, such as those described by Hinkel *et al.*[40] Figure 9 shows a quarter of the hohlraum in r-z space where the colormap corresponds to the product of the Coulomb logarithm and any correction terms based on the simulated plasma conditions and laser-intensity distribution at that time. The old default model in many codes used both the Lee and More Coulomb logarithm[8] and the standard Langdon absorption-reduction factor [Fig. 9(a)]. Keeping the plasma conditions fixed, the new model using Sommerfeld along with the new Langdon absorption-reduction factor as well as the ion-screening correction almost halves the absorption rate [Fig. 9(b)] (though naturally the changes will be more complex if the model is deployed inline in a self-consistent manner). This is expected to substantially alter the plasma conditions inside an indirect-drive hohlraum, which may help explain the long-standing inability to accurately predict implosion symmetry using the radiation-hydrodynamic simulations,[41,42] but a detailed study of the ramifications is left for future work.

VI. CONCLUSIONS

We have presented an updated analysis of recent measurements of inverse bremsstrahlung absorption through low-density gas-jet plasmas. Compared to the earlier publication,[4] plasma conditions were revised based on an improved understanding of the ionization states reached for the shots that included nitrogen, which was supported by a more careful analysis of the Thomson-scattering data as well as time-dependent Cretin simulations. Using also the densities and temperatures consistent with the new, lower ionization states, we were able to show that the model that best matches data uses the Sommerfeld Coulomb logarithm; an improved treatment of the Langdon effect, in which the Gaunt factor is properly integrated over the expected (and, in some cases, measured) non-Maxwellian, super-Gaussian electron distribution function; and an ion-screening correction factor with the screening length chosen to be the Wigner–Seitz radius.

We are most confident in the conclusions regarding the use of Sommerfeld for the Coulomb logarithm as well as the improved treatment for the Langdon effect, since they represent our best first-principles models. The screening correction factor, on the other hand, has the weakest theoretical foundation, and at the relatively low densities in these experiments, the fact that the screening correction reduced the discrepancy between predictions and measurements is merely suggestive but not definitive proof of the impact of screening. At the same time, this term extrapolates to having a very significant impact at the higher densities that dominate absorption in inertial confinement fusion, for example. In the future, we would like to see more theoretical and computational work studying the impact of screening on IBA in weakly coupled plasmas as well as experiments at higher density to further validate this effect.

Our ability to match measured absorption with calculations in most cases was of course made possible by the detailed plasma characterization provided by Thomson scattering. Applying this model to other platforms does not guarantee accurate results if the plasma conditions are not reasonably well known.

Note that we have not discussed the impact of laser speckles. The super-Gaussian exponents characterizing the EDFs were calculated using average intensities. However, Afeyan showed that intense speckles can locally enhance both the temperature and the super-Gaussian exponent, resulting in nonuniform plasma conditions.[43] Here, this effect was likely small because the absorption occurred over many speckles in both the transverse and longitudinal directions. It is possible, however, that our methodology could have masked this issue, since we measured the absorption of the Thomson probe beam that was also used to determine the plasma conditions—the probe necessarily samples the regions where its own speckles reside such that, for example, the inferred temperature was the most relevant temperature, even if slightly different from the true average temperature within the Thomson volume. For the moderate intensities employed here, the use of overlapping heater beams likely further smoothed the plasma conditions in the co-timed cases. At higher overlapped intensities outside the scope of this work, speckles might provide an extra source for collisions, modifying electron transport and potentially laser absorption.[44]

ACKNOWLEDGMENTS

We wish to acknowledge the help from A. Colaitis, D. H. Edgell, R. K. Follett, M. Foord, D. E. Hinkel, K. R. McMillen, and H. Scott. This material is based upon work supported by the DOE NNSA under Award No. DE-NA0003856, the University of Rochester, and the New York State Energy Research and Development Authority. This report was prepared as an account of work sponsored by an agency of the U.S. Government. Neither the

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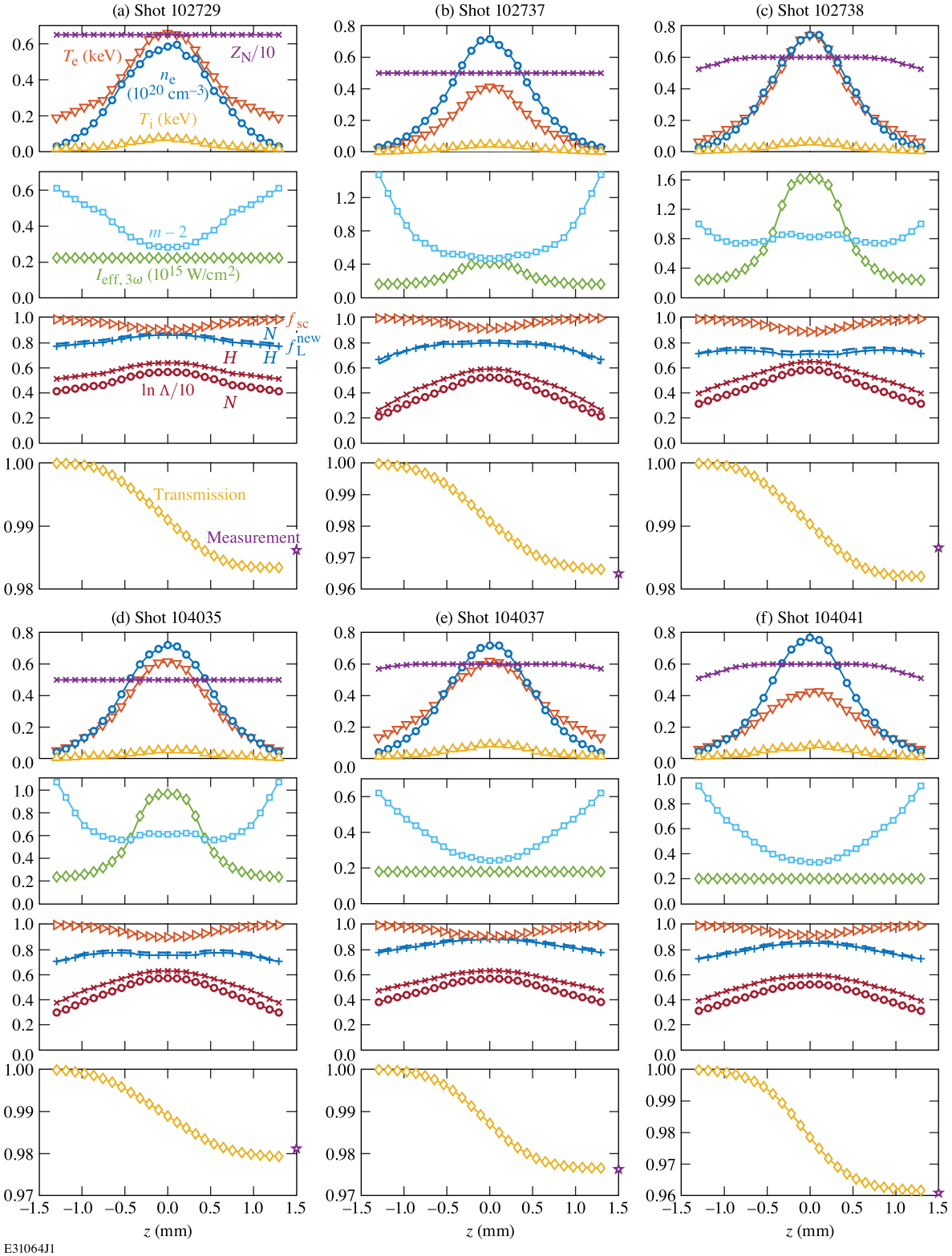


FIG. 10. N_2/H_2 shots 102729 – 104041.

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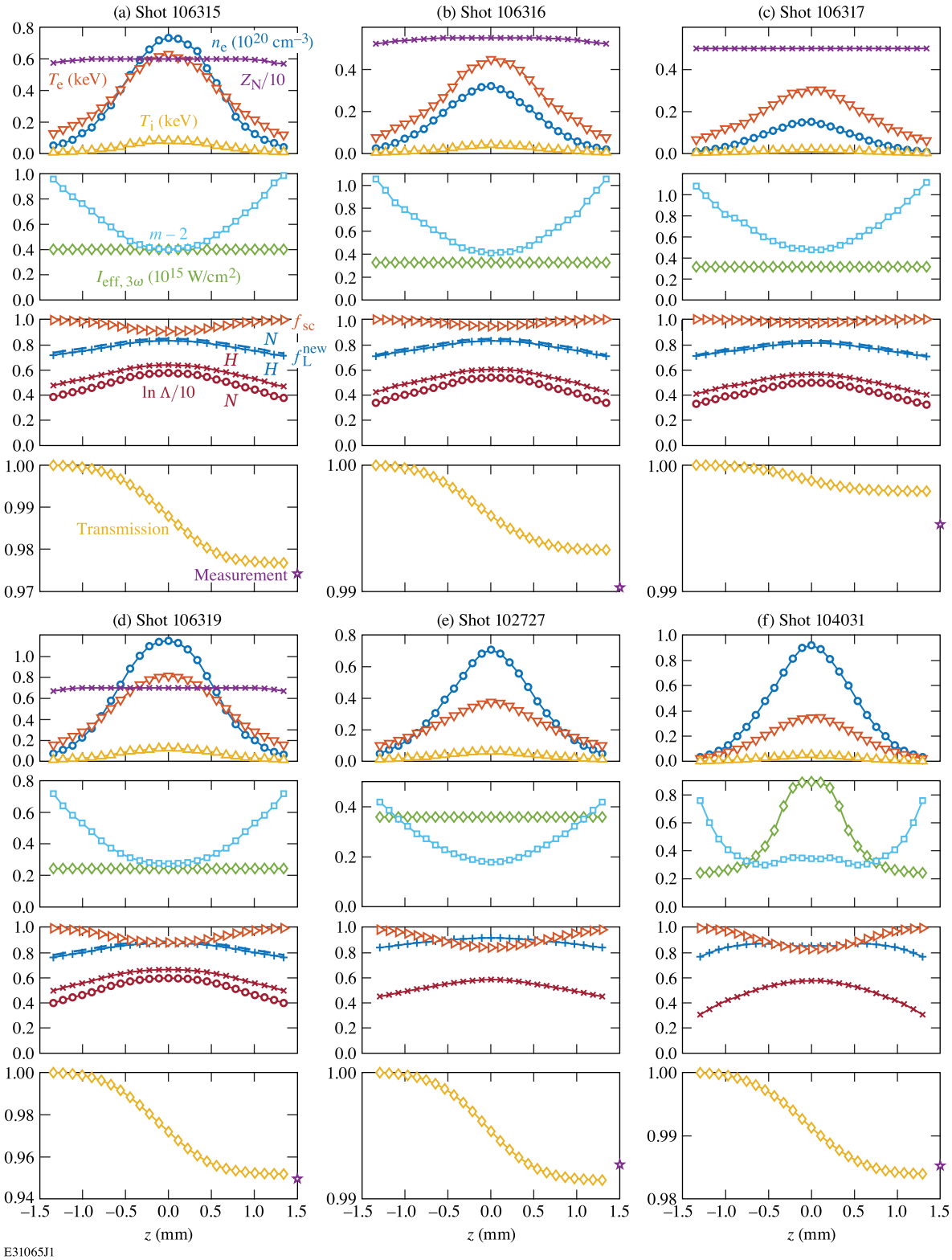


FIG. 11. N_2/H_2 shots 106315 – 106319 and H_2 shots 102727 – 104031.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

D. Turnbull: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **J. Katz:** Conceptualization (equal); Data curation (equal); Methodology (equal). **M. Sherlock:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Software (equal); Writing – review & editing (supporting). **A. L. Milder:** Conceptualization (equal); Formal analysis (supporting); Methodology (equal). **M. S. Cho:** Formal analysis (supporting); Software (supporting). **L. Divol:** Formal analysis (supporting); Methodology (supporting); Writing – review & editing (supporting). **N. R. Shaffer:** Conceptualization (supporting); Formal analysis (supporting); Writing – review & editing (supporting). **D. J. Strozzi:** Conceptualization (supporting); Formal analysis (supporting); Methodology (supporting); Writing – review & editing (supporting). **P. Michel:** Conceptualization (supporting); Methodology (supporting); Supervision (equal). **D. H. Froula:** Conceptualization (equal); Methodology (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon request.

APPENDIX A

Figures 10–12 show the shot data and results of the absorption calculations, which are provided here to assist those interested in testing alternative models as desired. In the second subfigure for each shot, note that m–2 is the deviation of the expected (and sometimes measured) super-Gaussian EDF from a Maxwellian. On those same plots, the green curve is the effective 3ω-equivalent intensity. {Recall this is the probe intensity only, multiplied by $(3/2)^2$, at late time, or the probe plus the overlapped heater-beam intensity at early time, and the Langdon parameter [Eq. (3)] is then calculated assuming 351-nm light.} In the third subfigure for each shot, the Coulomb logarithm(s) shown are based on the fit to Sommerfeld

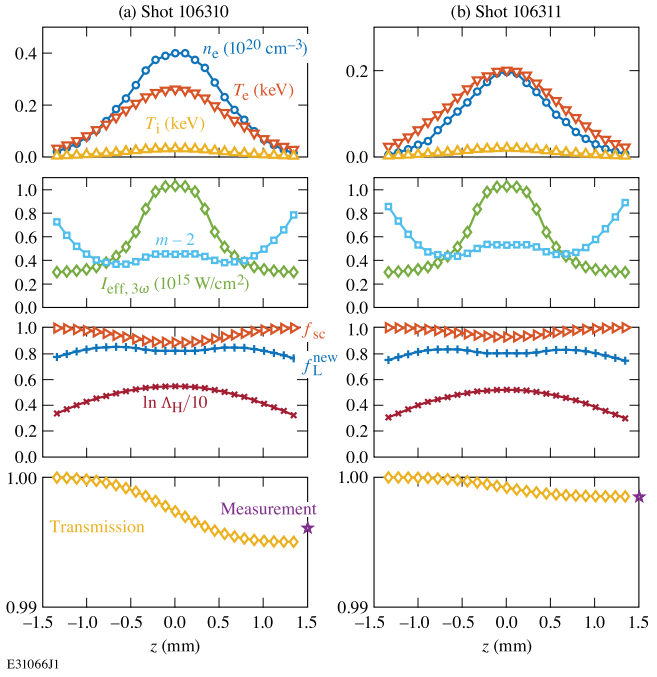


FIG. 12. H_2 shots 106310 – 106311.

[Eqs. (5), (7), and (8)]. The fourth subfigure shows the calculated transmission as a function of space using Eq. (4) and various inputs in the previous three subfigures, compared to the measured transmission for each shot.

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